Project Name: Green Lab Monitoring System

1. Introduction

A college laboratory is one of the most resource-intensive areas on campus, consuming significant amounts of **water, gas, and electricity** every day.

For laboratories to function efficiently, these utilities must be used responsibly and monitored continuously.

Currently, many colleges still depend on **manual supervision**, where lab assistants or faculty check whether:

- water taps are closed
- gas cylinders are safe
- electrical devices are switched OFF after use

This manual workflow often leads to:

- unnoticed leaks
- equipment left ON
- excessive electricity usage
- delayed maintenance
- increased operational cost

This project aims to solve these inefficiencies by transitioning from a **manual, reactive workflow** to a **smart digital monitoring system**.

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2. Stakeholders

The following groups are directly affected by this problem:

- **Students:** use the lab equipment daily and may unknowingly leave devices ON
- **Faculty Members:** need a safe and efficient lab environment
- **Lab Assistants:** responsible for daily monitoring and issue checking
- **College Administration:** wants reduced utility bills and sustainable campus operations
- **Maintenance Staff:** require accurate issue alerts and location-based repair tickets

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3. Problems Identified

Four core inefficiencies have been identified in the current lab environment:

Absence of Continuous Monitoring

There is no real-time system to track utility usage in laboratories.

Resource Wastage

Water leakage, gas leakage, and unnecessary electricity consumption often remain undetected for long periods.

Lack of Instant Alerts

Critical issues like gas leaks or equipment left ON are not reported immediately.

Absence of Historical Analytics

The administration lacks lab-wise reports to answer questions like:

"Which lab consumes the highest electricity every month?"

This prevents data-driven sustainability decisions.

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4. Existing System & Its Limitations

The current process relies heavily on **manual checking by staff**.

Process

- resources are used during practical sessions
- faculty or assistants manually inspect after classes
- issues are reported verbally

Limitations

This process is inefficient because:

- leaks may remain unnoticed
- no automated alerts exist
- no dashboard for usage comparison
- no cost analytics
- issue reporting is delayed
- no maintenance history is maintained

As a result, wastage and operational cost increase.

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5. Constraints & Assumptions

Time Constraint

The prototype must be designed and demonstrated within the **hackathon duration**.

Resource Constraint

The solution should use:

- low-cost sensors
- open-source backend technologies
- web-based dashboards

Deployment Assumption

The design assumes:

- laboratories have internet connectivity
- sensors can transmit real-time readings
- faculty/admin can access the dashboard using a modern browser

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6. One-Line Problem Summary

Traditional laboratories lack a smart monitoring system, leading to water, gas, and electricity wastage, delayed maintenance, and higher operational costs.